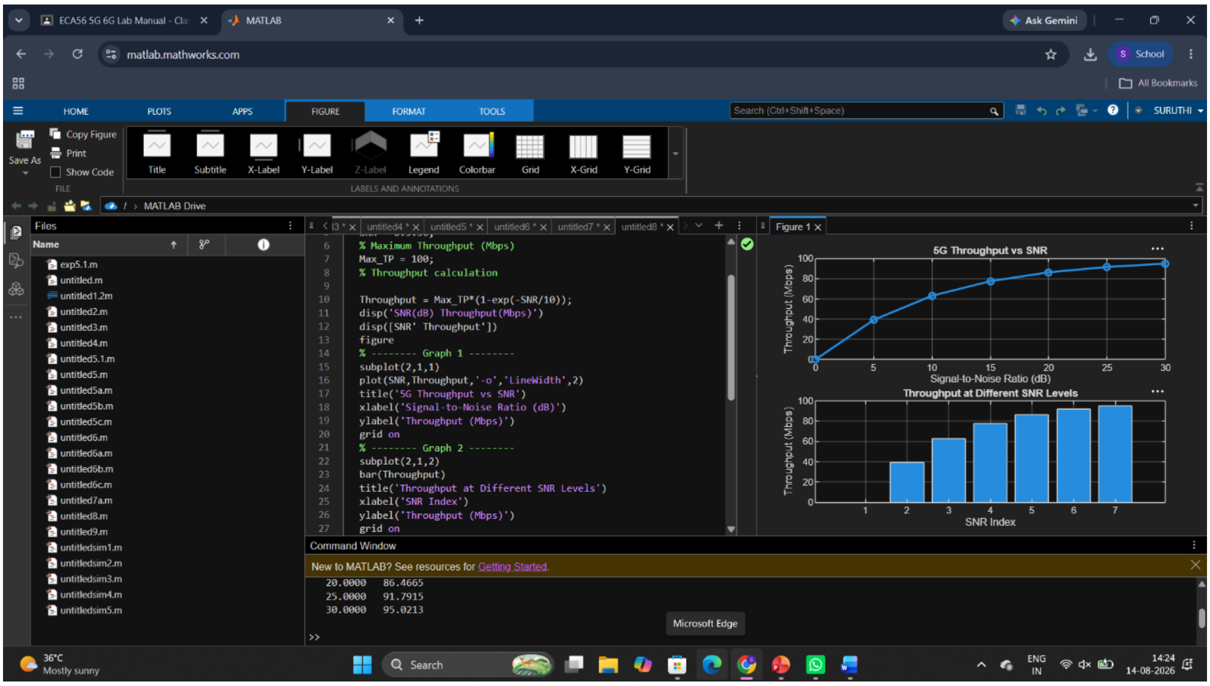
Ex.No. 14: 5G Throughput vs SNR with Modulation Labels

Objective1:

Program:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation
Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
…xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
bar(Throughput)
title('Throughput at Different SNR Levels')
xlabel('SNR Index')
ylabel('Throughput (Mbps)')
grid on
```

output:



Objective2:

Program:

clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Bit Error Rate (BER)

BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];

% Reliability (%)

Reliability = (1-BER)*100;

disp('SNR(dB) BER Reliability(%)')

…xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Bit Error Rate (BER)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)

plot(SNR,Reliability,'-s','LineWidth',2)

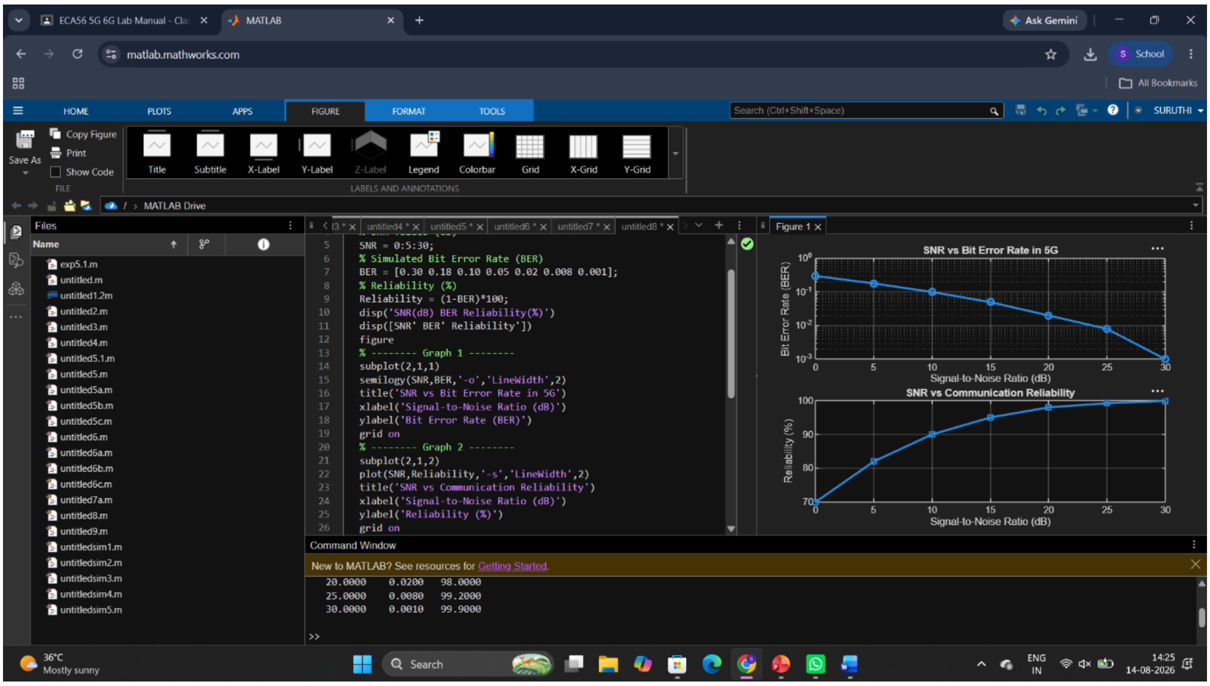
title('SNR vs Communication Reliability')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Reliability (%)')

grid on

output:



Objective3:

Program:

clc;

clear;

close all;

% Modulation schemes

Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};

% Required SNR (dB)

SNR = [5 10 18 25];

% Spectral Efficiency (bits/s/Hz)

Efficiency = [2 4 6 8];

```
disp('Required SNR (dB) and Spectral Efficiency')
…% -------- Graph 2 --------
subplot(2,1,1)
plot(1:4,SNR,'-s','LineWidth',2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')
grid on
```

output: